

Title Page

Long-term Clinical Outcomes of Ru-106 Plaque Brachytherapy for Choroidal Melanoma in Chinese Patients

Short title: Ru-106 Brachytherapy Outcomes for Choroidal Melanoma

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1 **Abstract**

2 Background and Objectives: This retrospective study aimed to summarize the clinical
3 characteristics, outcomes, and survival of Chinese patients with uveal melanoma
4 undergoing ruthenium-106 (Ru-106) plaque brachytherapy for the first time.

5 Materials and Methods: Clinical data from 117 patients treated at Peking University
6 People's Hospital between November 2008 and December 2019 were analyzed.
7 Univariate and multivariate regression analyses identified risk factors, and Kaplan-
8 Meier analysis calculated survival rates.

9 Results: The study cohort (mean age 48.4 years) predominantly comprised 66 men
10 and 51 women. Key findings included a median follow-up of 8 years, with a mean
11 largest basal tumor diameter of 18.3 mm and a mean largest tumor height of 7.0mm.
12 Local treatment failure occurred in 33.3% within 5 years, with metastasis in 21.4%,
13 mainly to the liver. Five-year and 10-year survival rates were 89.7% and 81.1%,
14 respectively. Tumor size and advanced tumor stage were significant predictors of
15 treatment failure, metastasis, and mortality.

16 Conclusions: Ru-106 brachytherapy showed promising outcomes in Chinese patients,
17 with younger onset and favorable survival rates compared to Western cohorts. Tumor
18 size was critical in predicting treatment success and survival. Ru-106 brachytherapy

19 appears beneficial for tumors with specific characteristics, potentially enhancing
20 survival and quality of life while minimizing ocular complications.

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35 **Introduction**

36 Uveal melanoma is the most common malignant intraocular tumor in adults,
37 affecting about 6 individuals per million per year in Europe and the United States [1].
38 In contrast, due to racial differences, its incidence in Asian populations is much
39 lower—estimated at 0.25 to 0.64 per million annually [2] [3].
40 Traditionally, enucleation was the standard treatment for choroidal melanoma.
41 However, advancements in medical technology have led to the development of eye-
42 conserving treatments such as laser therapy, radiotherapy, and surgical resection [4]
43 [5]. Several studies have revealed no clinically or statistically significant differences
44 in survival between patients treated with enucleation and brachytherapy [6] [7]. For
45 patients with a small- to medium-sized choroidal melanoma, these “conservative”
46 modalities have gradually become the preferred treatment options and be performed
47 individually or in combination with each other.
48 Brachytherapy, a form of radiotherapy, is typically administered using either iodine-
49 125 (I-125) plaques that emit gamma rays or ruthenium-106 (Ru-106) plaques that
50 emit beta rays. I-125 brachytherapy is widely used in North America, Australia, and
51 parts of China [8] [9] [10] [11], whereas Ru-106 is more commonly performed in
52 Europe [12] [13] [14]. These two modalities provide similar levels of tumor control,

enucleation rates, and patient survival. [15] [16] [17]. However, I-125 may be more effective for tumors with greater apical height. [16] [17]. Owing to its relatively high radiation exposure and deep penetration, I-125 brachytherapy is associated with significant visual deterioration in approximately 42–49% of patients within 3–4 years postoperatively [18]. In contrast, Ru-106 brachytherapy tends to preserve vision better and cause less tissue damage in patients with thinner melanomas because of their lower toxicity [15] [19].

Due to Asia's large and diverse population, significant opportunities exist to study choroidal melanoma [20] [21]. However, despite extensive research on brachytherapy in Western countries, long-term follow-up studies on Ru-106 radiotherapy in Asia are lacking. In this study, we focused on analyzing the clinical characteristics, changes in visual acuity, prognostic outcomes, 5-year and 10-year metastasis and survival rates in Asian patients with choroidal melanoma treated with Ru-106 radiotherapy. We also assessed the risk factors influencing their prognosis and outcomes.

Methods

Patients and data collection

The medical records of consecutive patients diagnosed with primary choroidal melanoma by the same ophthalmologist (LJH) and treated with Ru-106 brachytherapy

between November 2, 2008, and December 30, 2019 at the Ocular Oncology Center, Peking University People's Hospital, Beijing, were retrospectively reviewed. The exclusion criteria included tumor height outside 3.0–7.0 mm, prior treatment, bilateral tumors, planned resection following brachytherapy, other ocular surgeries, other malignancies, or lost to follow-up within the first three years. This study adhered to the Declaration of Helsinki and was approved by the Institutional Review Board, with informed consent waived due to deidentified data use. Among the 525 consecutive choroidal melanoma tumors diagnosed, 357 (68%) were treated with Ru-106 brachytherapy and 117 met the inclusion criteria. Patients underwent comprehensive ophthalmic exams, including best-corrected visual acuity (BCVA), intraocular pressure, slit-lamp biomicroscopy, ocular B-scan ultrasonography, fundus photography, fluorescein angiography, optical coherence tomography, and indirect ophthalmoscopy, conducted by senior ophthalmologists (LJH and CY). Additionally, patients received biochemical tests, liver/abdominal ultrasound, chest X-rays, and MRI every six months for two years, followed by annual imaging. Some also underwent abdominal CT. Patient demographics, tumor characteristics, treatment parameters, and outcomes regarding visual acuity, metastasis, and mortality were assessed during record and imaging review.

89 *Ruthenium plaque protocol*

90 Standard surgical techniques for ruthenium plaque brachytherapy were used. The
91 appropriate plaque—CCA, CCB, or COB (BEBIG GmbH, Germany)—was selected
92 based on the tumor's maximum basal diameter and location. The CCA and CCB
93 plaques measure 15.3 mm and 20.2 mm, respectively, whereas the COB plaque
94 features a groove to accommodate the optic nerve. After disinserting any muscle
95 overlying the tumor, its margins were identified via binocular indirect
96 ophthalmoscopy with indentation. The plaque was then positioned so that it extended
97 1–2 mm beyond the tumor margins in every direction. Sutures secured the plaque's
98 lugs to the sclera, with additional mattress sutures ensuring firm attachment to the
99 globe. Any disinserted muscles were reattached using slings anchored to the sclera,
100 plaque lugs, or sutures. Treatment times were calculated based on tumor apical height,
101 following the manufacturer's specifications. Typically, a minimum scleral dose of
102 300–400 Gy and an apex dose of 80–100 Gy were administered, with a 5–10%
103 margin.

104 *Follow Up*

105 Patient survival status was determined through outpatient records; telephone
106 interviews with the patient, their contacts, or local hospitals; and personal visits to the

patient or family. Follow-up outcomes were categorized as melanoma-free survival, metastasis, melanoma-related death, or death from other causes.

Outcome measures

Demographic data, clinical characteristics, treatment-related complications, and tumor-related data were collected from patients' medical records and imaging studies.

Patient Characteristics

Demographic data encompassed age, sex, and medical history. Clinical characteristics included BCVA, lens status, and anterior segment abnormalities. BCVA was measured using a Snellen chart, with pinhole correction as needed. The baseline visual acuity (VA) was the BCVA measured closest to the time before brachytherapy; the follow-up VA was the BCVA recorded during clinical reviews; and the final VA was the BCVA at the last follow-up. Treatment-related complications encompassed various forms of radiation ophthalmopathy, including new-onset radiation-induced maculopathy, retinopathy, optic neuropathy, and cataract; other ocular surface disorders; ischemic retinopathy; neovascularization; neovascular glaucoma; and vitreous hemorrhage.

Tumor Characteristics

Tumor data included the location of the tumor epicenter (iris, ciliary body, or

choroid), tumor location classification (anterior to the equator, equatorial, posterior to the equator, or close to the optic nerve), tumor quadrant (macular, nasal, temporal, superior, superior nasal, inferior nasal, superior temporal, or inferior temporal), and anterior/posterior tumor margins (ciliary body, equator to ora serrata, posterior to equator). Distances to the proximal edge of the optic disc and foveal avascular zone (mm) were noted. Tumor characteristics included maximum basal diameter and apical height via B-scan ultrasound, and features like subretinal fluid, macular edema, pigmentation grade, Bruch's membrane rupture, extraocular spread, and vitreous hemorrhage. Tumor size and shape were classified per the Collaborative Ocular Melanoma Study (COMS) criteria: small (< 2.5 mm maximum apical height and/or ≤ 8 mm maximum basal diameter), medium (2.5–10 mm maximum apical height and/or 8–16 mm maximum basal diameter), and large (> 10 mm maximum apical height and/or ≥ 16 mm maximum basal diameter). Tumor shapes were categorized as dome, mushroom, peaked, flat, or irregular. The 8th edition of the American Joint Committee on Cancer (AJCC) criteria was used for TNM classification.

Treatment Characteristics

Measured brachytherapy parameters included the Ru-106 plaque type, radiotherapy duration (hours), and doses delivered to the tumor apex and sclera (Gy). Local

treatment failure was identified by either a $\geq 15\%$ increase in tumor height via B-scan ultrasound or a 250 μm expansion of any tumor boundary, as assessed by fundus photography or clinical ophthalmoscopy. Upon brachytherapy failure, ocular oncology specialists opted for local resection or enucleation based on the tumor characteristics and the patient's systemic condition, with ongoing monitoring. Indications for local resection included tumor enlargement post-brachytherapy or dispersion of necrotic tumor tissue into the vitreous cavity. If disease progression occurred after tumor resection, enucleation was performed.

During follow-up, systemic examinations were conducted by relevant specialists. In cases of systemic metastasis, the time to metastasis and any melanoma-related death were documented.

Statistical Analysis

Descriptive statistics were used to summarize the baseline characteristics, visual acuity changes, radiotherapy details, radiological complications, and local tumor control, expressed as means $\pm$ standard deviations. VA measurements were converted to logarithms of the minimum angle of resolution (logMAR) units for analysis. Five- and ten-year metastasis rates and survival were evaluated, with survival time defined as the interval from initial treatment to death or last follow-up. Kaplan-Meier

estimates were used to plot survival curves, with the primary endpoint for metastasis-free survival being the time to metastatic disease development, treating deaths from other causes as censored events. Subgroup analyses were performed based on metastasis status, patient survival, and local treatment failure. Univariate regression analysis of baseline demographics, tumor features, and VA changes identified factors predicting treatment response ($p < 0.20$). TNM stage was selected as a potential predictor in multivariate linear regression using a stepwise forward procedure. Multivariate linear regression assessed interactions between primary outcomes and each factor. Statistical analyses were conducted using IBM SPSS (Version 27.0, IBM Corporation, Armonk, NY, USA) and GraphPad Prism (Version 8.0.2, GraphPad Software, Inc., CA, USA) software, with a P value < 0.05 indicating statistical significance.

Results

From 2008 to 2019, 117 eligible patients (56.4% male) were included, with a median age at diagnosis of 50 years (range 18–79) and an average follow-up of 8.6 ± 3.4 years. None of the patients had prior tumors. Twelve patients (10.3%) had hypertension, and all were phakic at diagnosis. Baseline ocular abnormalities included mild (46 patients, 39.3%) or moderate (49 patients, 41.9%) cataracts, iris

neovascularization (3 patients, 2.6%), macular edema (15 patients, 12.8%), retinal detachment over the tumor (38 patients, 32.5%), and Bruch's membrane rupture (5 patients, 4.3%). During brachytherapy, 23 patients (19.7%) developed radiation-induced eye diseases after 3.3 ± 1.2 years, including radiation-induced cataracts (16 patients), maculopathy (9 patients), radiation optic neuropathy (7 patients), radiation retinopathy (4 patients), and neovascular glaucoma (16 patients). Among them, sixteen patients had multiple radiation-related ocular conditions.

At baseline, 22.2% of patients had a VA of 20/40 or better in the affected eye, with a mean logMAR VA of 0.75 ± 0.64 (median 0.6; range 24/20 to light perception). Only 16.2% had a VA of 20/200 or worse. At the 3-year follow-up, the mean VA declined to logMAR 1.00 ± 0.68 . Specifically, 15.4% patients had a VA between 20/25 and 20/40, 53.0% between 20/50 and 20/200, and 31.6% of 20/200 or worse.

Based on COMS tumor size classification, 1 tumor (0.9%) was small, 106 (90.6%) medium, and 10 (8.5%) large. Tumors had an average maximum basal diameter of 10.8 ± 2.8 mm (range 4.4–18.3 mm) and an average apical height of 5.5 ± 1.3 mm (range 3.1–7.0 mm); the changes are shown in Figure 1. All 117 tumors were completely or partially in the choroid, with 14 (12%) extending anteriorly to involve the ciliary body. The tumors were most commonly located in the superior region

(19.7%), followed by the macular region (14.5%), temporal (18.0%), superior-temporal (11.4%), inferior-temporal (8.4%), inferior (6.6%), nasal (6.6%), inferior-nasal (6.4%), and superior-nasal (5.7%). Mean distances to the optic disc and fovea were 3.9 ± 3.1 PD and 3.6 ± 3.1 PD, respectively. According to the COMS shape classification, 34 tumors (29.1%) had a dome, 28 (23.9%) a mushroom, 10 (8.6%) a peaked, 24 (20.5%) a flat, and 21 (17.9%) an irregular shape. TNM staging classified 19 patients as stage I, 89 as stage II (57 IIA, 32 IIB), and 9 as stage III, with no cases of baseline metastasis.

CCB plaques were used with 77 patients, COB with 19, and CCA with 21 (Table 2). The Ru-106 plaques delivered a median scleral dose of 650 Gy (mean 616.8 ± 185.5 Gy) and a median apex dose of 100 Gy (mean 96.5 ± 15.3 Gy). Figure 2 shows the pretreatment and posttreatment appearance of a representative case.

The follow up outcomes in Table 3 revealed brachytherapy failure in 39 eyes (33.3%) within five years, with a mean time to failure of 1.06 ± 1.49 years. Upon confirmation, ophthalmologists performed enucleation on 19 eyes and local resection on 20 eyes according to the ocular condition and patient requirements. Among them, five eyes initially managed by local resection later required enucleation because of the failure of eye-preserving treatment, with a mean interval of 1.52 ± 1.32 years. Over 80% of

failures occurred within five years, while only eight eyes underwent resection after five years; among these, two eventually had enucleation for reasons unrelated to treatment failure. Most enucleations were due to tumor growth, with some resulting from pain, poor vision, or other causes (Supplementary Table 1). During the study period, 18 of 117 patients (15.4%) died, including 16 patients who died of melanoma-related metastasis and 2 patients who died of cirrhosis and myocardial infarction. Tumor metastasis was detected in 25 patients (21.4%), with an average time to metastasis of 3.81 ± 1.96 years. The 5-year and 10-year metastasis rates were 17.1% and 28.3%, respectively. Patients treated with brachytherapy alone had a lower metastasis rate, with 5-year and 10-year metastasis rates of 16.4% and 21.4%, and an earlier average time to metastasis of 2.77 ± 1.84 years. At final follow up, 90 patients (76.9%) were alive without metastasis or recurrence (Table 3). Systemic metastasis of choroidal melanoma most commonly involved the liver (13 cases, 52%), followed by the lung (6 cases, 24%) and bone (2 cases, 4%) (Supplementary Figure 1). The average time from metastasis to death was 1.25 ± 0.78 years (range 0.3–3 years) with a 1-year survival rate of 22.22%. For brachytherapy-only patients, the average time from metastasis to death was slightly longer at 1.32 ± 0.86 years (range 0.2–2.8 years), with a 1-year overall survival rate of 23.08%.

233 Among all 117 patients, the 5- and 10-year survival rates were 89.7% and 81.1%,
234 versus 83.6% and 75.0% in the brachytherapy-only group (78 patients). No significant
235 difference in 10-year survival was observed between the groups ($p = 0.07$) (Figure 3).
236 Univariate analysis linked 5-year treatment failure risk to larger tumor size, shorter
237 tumor-to-optic disc distance, and lower apex radiation dose (all $p < 0.05$). Greater
238 apical height and poorer visual acuity at baseline were marginal (all $p < 0.20$).
239 Multivariate analysis identified only larger tumor size and closer optic disc distance as
240 independent predictors (all $p < 0.05$) (Supplementary Figure 2). Univariate analysis
241 showed poorer visual acuity and greater tumor width and height at baseline were
242 associated with tumor metastasis ($p < 0.05$); older age and lower apex radiation dose
243 were marginal ($p < 0.20$). Multivariate analysis revealed only poorer visual acuity,
244 greater tumor height at baseline, and advanced T stage independently predicted tumor
245 metastasis ($p < 0.05$) (Supplementary Figure 3). Univariate analysis linked
246 melanoma-related death to earlier metastasis and greater baseline tumor width and
247 height ($p < 0.05$). Marginal associations included lower apex radiation dose, poorer
248 visual acuity, shorter tumor-optic disc distance, larger tumor size, and 5-year
249 treatment failure ($p < 0.20$). Multivariate analysis identified greater apical height,
250 earlier metastasis, and advanced T stage as predictors ($p < 0.05$) (Supplementary

251 Figure 4).

252 Discussion

253 This first large-scale single-center study of Ru-106 brachytherapy for uveal
254 melanomas in Asia, with a follow-up period exceeding 10 years, found an average
255 onset age of 48.4 years and a roughly equal sex distribution, aligning with previous
256 findings that Asians are younger at diagnosis than Caucasians [11] [22] [23].

257 Although Western studies indicate that uveal melanomas have a higher incidence in
258 males [24], this finding has not been confirmed in the Asian population. Notably, our
259 univariate analysis showed male sex was strongly associated with tumor-related death
260 ($p < 0.05$), although sex differences were ultimately excluded from the multivariate
261 analysis. Some studies also report that sex differences are related to patient survival
262 rates [25] [26], showing the risk of metastatic uveal melanoma-related death is
263 significantly higher in males than in females [27].

264 Using the definition put forward in the COMS, we classified tumors by largest basal
265 diameter and maximum apex height. Medium-sized tumors were most common
266 (90.6%), followed by large, with small tumors being rare. The average basal diameter
267 was 10.8 ± 2.8 mm (range 4.4–18.3 mm) and the average apex height was $5.5 \pm$
268 1.3 mm (range 3.1–7.0 mm). Prior studies have revealed that Asians have larger basal

tumor diameters than Caucasians, ranging from 12 mm to 14.8 mm [23] [28] [29].

The size of the tumor has long been considered one of the most important clinical prognostic features of uveal melanoma [30] [31]. Our multivariate linear regression analysis found tumor size (including basal width and apex height) correlated with local treatment failure, metastasis, and melanoma-related death—confirming tumor size as a key prognostic factor, consistent with findings by Kujala et al. and Shields et al [32] [33].

Currently, no criteria exist for selecting the optimal plaque size and type for choroidal melanoma. While some researchers advocate Ru-106 for smaller tumors [34] [15], we attempt to maximize its use for eye preservation. In this study, we treated 51 patients (43.6%) with tumors having an apex height of 6.0–7.0 mm, with a mean follow-up of 9.1 ± 3.3 years. Among these, 9 patients (45.0%) underwent intraocular resection and 5 (25.0%) enucleation after an average of 1.1 ± 1.5 years due to treatment failure. Additionally, 12 patients (23.5%) developed metastasis after 4.1 ± 1.9 years, and 6 (11.8%) died on average 2.1 ± 1.0 years post-metastasis. In summary, we found that, of all the patients included in this study, those treated with Ru-106 radioactive plaques for tumors with apex heights between 6.0–7.0 mm had similar average times to treatment failure and systemic metastasis. Does this result suggest the feasibility of

using Ru-106 radioactive plaques for patients with greater tumor heights but a strong need for eye preservation? Although further evidence is needed to draw definitive conclusions, our findings offer a new perspective on eye-preserving treatment strategies for patients with large choroidal melanomas.

Studies report radiation-induced eye disease rates of 25–60% in choroidal melanoma patients [35] [36] [37], with over 40% suffering severe vision loss (worse than 20/200) within three years post-brachytherapy. In our study, radiation ophthalmopathy occurred in 19.7% of patients, with vision worse than 20/200 increasing from 16.2% at baseline to 31.6% within three years after the procedure, requiring intervention in only 20 cases. This lower complication rate may be attributed to using Ru-106 plaques and restricting treatment to tumors with apex heights of 3–7 mm.

The 5-year local treatment failure rate was 39.3%, contrasting with the COMS's 90% tumor control [6]. Owing to differences in study design, patient selection, follow-up, definitions of tumor growth or local failure, and analysis complicate comparisons, it is difficult to compare our results with those of other studies. Moreover, adjusted tumor size indications in this study for Ru-106 may have contributed to the higher local treatment failure rate.

Notably, in our study, overall 5-year and 10-year melanoma-related mortality rates were 10.3% and 18.9%, respectively, aligning with COMS medium-sized tumors treated with I-125 brachytherapy [38]. However, patients treated with Ru-106 alone had higher melanoma-related mortality rates of 16.4% and 25%, falling between the COMS rates for large (28% and 40%) and medium-sized tumors [39]. This may be due to the tumor sizes in this study generally being between those in the two subgroups in the COMS. Asian choroidal melanoma patients have slightly higher survival rates compared to previous reports. A study of 5788 choroidal melanoma patients in Europe revealed a 5-year relative survival of 69% [40], and Bakalian noted 65% at 5 years and 52% at 10 years [41]. Additionally, Maria Fili et al. treated 571 patients with tumors ≥ 5.5 mm using Ru-106 or I-125 plaques, finding 10-year survival rates of 59% (Ru-106) and 56% (I-125) for tumors 5.5–7.4 mm, and 46% (Ru-106) and 44% (I-125) for tumors ≥ 7.5 mm, with no significant difference among different plaque types [16]. In Asian populations, the 5-year and 10-year survival rates of choroidal melanoma range from 70% to 90% [11] [23] [42] [43]. Although the radiation brachytherapy methods selected and the inclusion criteria differ across studies, the survival rates differ significantly from those in Western populations. In addition to differences in genetics and the age at onset among different ethnic groups [23] [44] [45] [46], many studies have revealed that microRNAs are potential

quantitative biomarkers for the diagnosis and prognosis of uveal melanoma [47] [48].

Further research with larger groups of patients and diverse ethnic populations is

needed to confirm these findings.

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Figure legends

531 Figure 1: Changes in (a) apical height and (b) basal diameter of tumors in each of the
532 117 patients.

533 Figure 2: (a) and (b) Left fundus of a 51-year-old woman before brachytherapy, with
534 a B-scan ultrasound showing an inferior temporal choroidal melanoma measuring
535 6.13 mm in height and 12.76 mm in width with eccentric plaque placement. The
536 tumor extended close to the fovea, reducing her visual acuity to 20/40. (c) and (d) Six
537 months after treatment, the tumor height had decreased to 4.21 mm, and the width had
538 decreased to 11.14 mm. Her visual acuity improved to 20/33.

539 Figure 3: Kaplan–Meier estimates of survival rates of uveal melanoma patients.

Table 1 Choroidal Melanoma Patient Demographics and Tumor Characteristics.

Characteristic	No.	%
Demographic characteristics		
Gender		
Male	66	56.4
Female	51	43.6
Age		
< 40	29	24.7
40-59	65	55.6
≥ 60	23	19.7
Hypertension		
Yes	12	10.3
No	105	89.7
Ophthalmic characteristics		
Visual acuity in eye with tumor in baseline		
≥ 20/20	5	4.3
20/25-20/40	26	22.2
20/50-20/200	67	57.3
≤ 20/200	19	16.2
NA	0	0
Visual acuity in fellow eye in baseline		
≥ 20/20	73	62.4
20/25-20/40	25	21.4
20/50-20/200	17	14.5
≤ 20/200	2	1.7
NA	0	0
Lens status in eye with tumor		
Phakic	117	100
Aphakic or pseudophakic	0	0

Degree of cataract			
None	22	18.8	
Mild	46	39.3	
Moderate	49	41.9	
Severe	0	0	
Abnormal eye manifestations			
Iris neovascularization			
Yes	3	2.6	
No	104	97.4	
Macular edema			
Yes	15	12.8	
No	102	87.2	
Retinal detachment over tumors			
Yes	38	32.5	
No	79	67.5	
Bruch's membrane			
Unbroken	83	70.9	
Indeterminate	29	24.8	
Broken	5	4.3	
Tumor characteristics			
Tumor apical height (mm)			
3.0-4.0	26	22.2	
4.1-6.0	40	34.2	
6.1-7.0	51	43.6	
Longest basal dimension (mm)			
4.5-8.0	13	11.1	
8.1-11.0	51	43.6	
11.1-14.0	42	35.9	
14.1-18.5	11	9.4	
Tumor size grading			

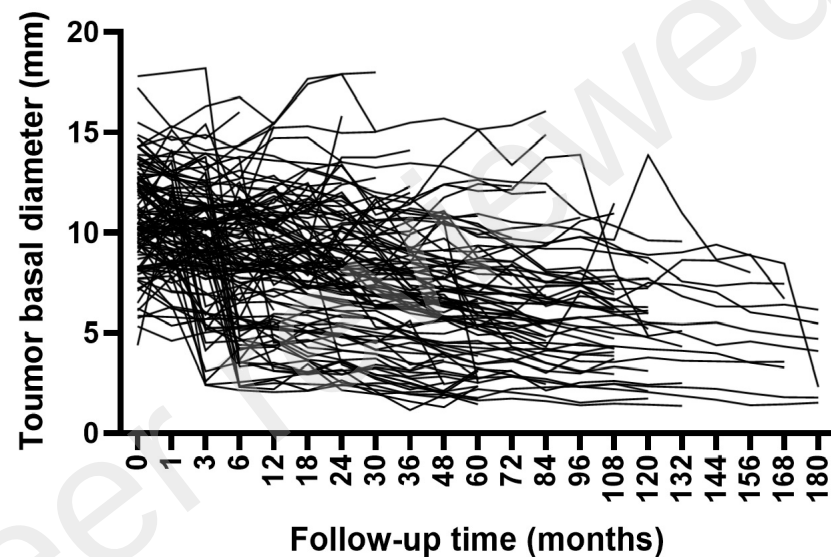
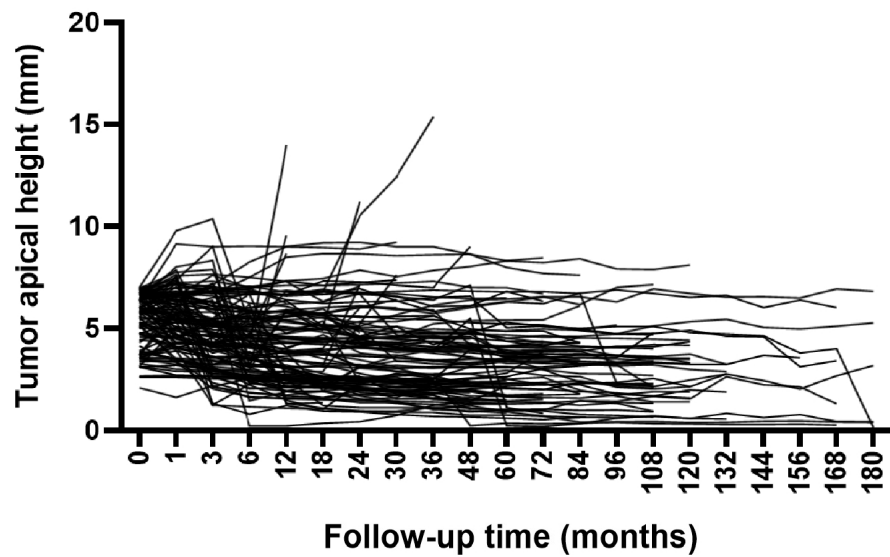
Small	1	0.9
Medium	106	90.6
Large	10	8.5
Tumor TNM classification		
Stage I	19	16.2
Stage II	89	76.1
Stage IIA	57	64
Stage IIB	32	36
Stage III	9	7.7
Tumor metastasis		
Yes	0	0
No	117	100
Location of anterior tumor border		
Ciliary body	14	12.0
Equator to ora serrata	81	69.2
Posterior to equator	22	18.8
Location of posterior tumor border		
Equator to ora serrata	15	12.8
Posterior to equator	102	87.2
Tumor epicenter		
Iris	0	0
Ciliary body	0	0
Choroid	117	100
Tumor quadrant		
Macular	17	14.5
Nasal	10	8.5
Temporal	15	12.8
Superior	23	19.7
Superior- Nasal	7	6.1
Superior- Temporal	12	10.2

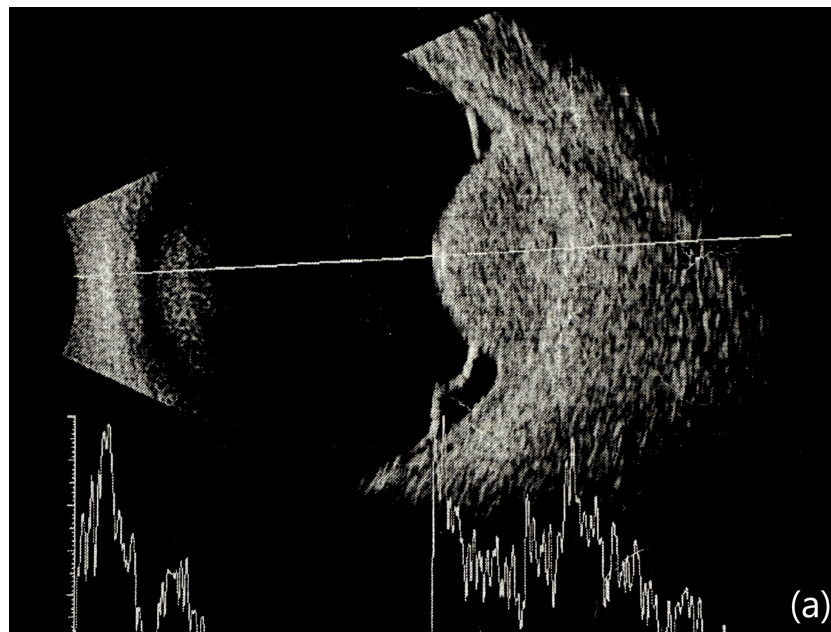
Inferior	11	9.4
Inferior- Nasal	10	8.5
Inferior- Temporal	12	10.3
Tumor location classification		
Anterior to the equator	11	9.4
Equatorial	33	28.2
Posterior to the equator	71	60.7
Close to optic nerve	2	1.7
Distance to proximal edge of optic disc (mm)		
≤ 2.0	34	29.1
2.1-4.0	30	25.6
4.1-6.0	27	23.1
6.1-8.0	17	14.5
> 8.0	7	6.0
NA	2	1.7
Distance to center of foveal avascular zone (mm)		
0.0	23	19.7
0.1-2.0	21	17.9
2.1-5.0	40	34.2
5.1-8.0	22	18.8
> 8.0	11	9.4
Tumor shape		
Dome	34	29.1
Mushroom	28	23.9
Peaked	10	8.6
Flat	24	20.5
Irregular	21	17.9
Grade of pigmentation		

Light to amelanotic	26	22.2
Medium	43	36.8
Heavy	41	35.0
Mixed color	7	6.0

Table 2 Treatment characteristics of patients with choroidal melanoma.

Treatment Characteristic	No.	%
Types of plaque brachytherapy		
CCB	77	65.8
COB	19	16.2
CCA	21	17.9
Treatment time (hr)		
≤ 72	23	19.7
73-144	56	47.8
> 144	38	32.5
Dose to tumor apex (Gy)		
< 80	6	5.1
80-100	66	56.4
101-120	23	19.7
125-150	22	18.8
Dose to sclera (Gy)		
< 300	2	1.7
300-349	13	11.1
350-399	7	6.0
400-499	11	9.4
500-599	10	8.5
600-699	23	19.7
700-799	27	23.1
≥ 800	24	20.5





(a)



(c)



(b)



(d)

Strata  Treatment=Brachytherapy with enucleation or local resection  Treatment=Only brachytherapy

